

ORIGINAL ARTICLE

Expected ranges for blood neutrophil concentrations of neonates: the Manroe and Mouzinho charts revisited

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Background: Upper and lower reference limits for blood neutrophil concentrations of neonates were reported by Manroe, Rosenfeld and co-workers in 1979 and by Mouzinho, Rosenfeld and co-workers in 1994. Studies by Carballo and co-workers and Maynard and co-workers suggested that at high altitude a much higher upper limit of values occurs, and that neutrophilia is often diagnosed inappropriately, at high altitude, if the Manroe and Mouzinho charts are used.

Study Design: We revisited the Manroe and Mouzinho charts using modern cell counting instrumentation and very large sample sizes. This was accomplished with data from an 18-hospital health-care system, at hospitals averaging 4800 feet above sea level. Data were obtained from neonates in the first 10 days following birth, using tests performed on a single type of automated blood cell counter (Beckman Coulter LH 750, Fullerton, CA, USA). To obtain a description of 'expected values,' patients with the following diagnoses, known to have a high proportion of abnormal neutrophil counts, were excluded from the analysis; (1) maternal pregnancy-induced hypertension, (2) early-onset bacterial sepsis, (3) a discharge diagnosis of congenital neutropenia and (4) trisomy 21, 18 or 13.

Result: Blood neutrophil concentrations were tabulated from 30 354 tests performed on neonates of 23 to 42 weeks gestation, with dates of birth between 1 January 2004 and 31 May 2007. The reference range charts generated from these data have much higher upper value limits than do the Manroe and Mouzinho charts, and are similar to the high-altitude reports of Carballo and co-workers and Maynard and co-workers. Neonates whose mothers labored before delivery had significantly higher neutrophil counts (averaging 3500 neutrophils per μl higher, $P < 0.0001$; comparing cesarean section deliveries with vs without labor). Females had significantly higher counts (averaging 2000 neutrophils per μl higher) than males ($P < 0.0001$). Counts did not differ on the basis of racial/ethnic group.

Conclusion: The figures of this report describe the expected ranges for blood neutrophil concentrations over the first 10 days of life among neonates of 23 to 42 weeks gestation at high altitude.

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Introduction

In 1979, Manroe, Rosenfeld and co-workers¹ from the University of Texas Southwestern Medical School in Dallas published reference values for blood neutrophil concentrations in neonates. That landmark publication established a method for determining whether a neonate's neutrophil concentration should be considered neutrophilic, normal or neutropenic. Figure 1 of the Manroe publication¹ displayed the expected range of neutrophil concentrations during the first 60 h after delivery. The data were obtained from 108 neonates hospitalized during the years 1974 to 1976. The complete chart (Figures 1 and 2 of the Manroe publication¹), extending to 700 h after delivery, was constructed from blood tests obtained from 434 neonates with a gestational age of 38.9 ± 2.4 weeks (mean $\pm$ s.d.).

Few very low birth weight (<1500 g) neonates were included in the 1979 Dallas publication, although no differences in neutrophil concentrations were observed between those less than vs greater than 2000 g birth weight.¹ However, in 1994 the same group (Mouzinho *et al.*²) published new information on neutrophil concentrations specifically from very low birth weight neonates. They observed that the range of counts differed from that of higher gestation neonates, with a wider range of distribution. Sixty-three counts, including only neonates <1500 g birth weight, which had no recognized complications that would modify their neutrophil counts formed the basis of the chart describing the first 60 h after delivery. One hundred thirty-one counts were used for the values beyond 60 h.² The Mouzinho report has subsequently been widely used to assess the normalcy of neutrophil counts from very low birth weight neonates, while the original Manroe report¹ is used to

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assess the normalcy of neutrophil counts from term and near-term neonates.³

During the past decade, new automated blood cell counting instrumentation has been introduced into hospital practice, with many advantages over previous methods.⁴ One advantage is the considerably higher number of cells sampled to generate the report. Another advantage is elimination of technician-to-technician variance inherent in the visual identification of neutrophils.⁴ This study was accomplished to take advantage of modern methods of neutrophil quantification, and to tabulate data from the very large sample size of an 18-hospital health-care system, Intermountain Healthcare. Using recent data obtained from a single model of electronic cell counter, (Beckman Coulter LH750) we revisited the Manroe¹ and Mouzinho² charts, describing the normal range of blood neutrophil concentrations of neonates in our health-care system.

Methods

Data were collected electronically as a deidentified limited data set from archived Intermountain Healthcare reports. The information collected was limited to that displayed in this report. Neutrophil concentrations (neutrophils per μl blood) were obtained if the patient had a date of birth from 1 January 2004 through 31 May 2007, was cared for in any Intermountain Healthcare facility, and had a complete blood cell count (CBC) run on a Beckman Coulter LH750. The values were included in the analysis only if the CBC was obtained within the first 10 days (240 h) after delivery. Results were excluded from patients with the following diagnoses: high proportion of abnormally low neutrophil concentrations; maternal pregnancy-induced hypertension;^{5,6} early-onset bacterial sepsis^{7,8} or a discharge diagnosis of congenital neutropenia;^{9,10} or abnormally high neutrophil concentrations; Down syndrome;¹¹ trisomy 13¹² or trisomy 18;¹² or early-onset bacterial sepsis.¹³ The gestational ages used in this study were those assigned by obstetrical dates, unless this was changed by the neonatal examination.

The program used for data collection was a modified subsystem of 'clinical workstation.' Clinical workstation is a web-based electronic medical record application that stores demographic and clinical information, such as history, physical examination results, laboratory data, problem lists and discharge summaries. 3M Company (Minneapolis, MN, USA) approved the structure and definitions of all data points for use within the program. The data were collected from the electronic medical record, laboratory systems and case mix. Case mix is the billing, coding and financial data mart used by Intermountain Healthcare. Data were managed and accessed by authorized data analysts. The Intermountain Healthcare Institutional Review Board approved the study. Descriptive statistics were calculated using Statit (Corvallis, OR, USA). Means and 5th percentile and 95th percentile intervals were used to express the neutrophil concentrations. Differences in

neutrophil concentrations between groups were assessed using Student's *t*-test. *P*-values <0.05 were considered significant.

Results

A total of 30 354 CBCs were included in this report. Blood neutrophil concentrations, obtained in the first 72 h following birth of neonates >36 weeks gestation, are displayed in Figure 1. Figures 2 and 3 display neutrophil concentrations in the same way, using data from patients 28 to 36 weeks gestation in Figure 2, and <28 weeks gestation in Figure 3. During the first 3 days, neutrophil concentrations were lowest at the earliest times sampled, including umbilical cord (time 0). Counts increased over the first hours following delivery, with peak values occurring between 6 and 8 h for those ≥ 28 weeks gestation, but at 24 h for those delivered at <28 weeks gestation.

In the interval between 72 and 240 h, the 5th percentile value for blood neutrophil concentrations of neonates >36 weeks gestation was $2700 \mu\text{l}^{-1}$ and the 95th percentile value was $13\,000 \mu\text{l}^{-1}$. During this period (72 to 240 h), for neonates born at 28 to 36 weeks gestation the 5th percentile value for blood neutrophil concentration was $1000 \mu\text{l}^{-1}$ and the 95th percentile value was $12\,500 \mu\text{l}^{-1}$. For neonates born <28 weeks gestation, the 5th percentile value for blood neutrophil concentration was $1300 \mu\text{l}^{-1}$ and the 95th percentile value was $153\,000 \mu\text{l}^{-1}$.

The concentrations of immature neutrophil per μl blood, during the first 72 h following delivery of neonates >36 weeks gestation, are shown in Figure 4. The highest concentrations were generally present closest to delivery, with a gradual fall from about 12 h through the first 3 days.

During the first 18 h following delivery, neonates whose mothers labored had significantly higher neutrophil counts (Figure 5). Specifically, neonates delivered by cesarean section following labor had an average neutrophil concentration of $12\,020 \mu\text{l}^{-1}$ compared with $8650 \mu\text{l}^{-1}$ among those born by cesarean section with no preceding labor ($P < 0.0001$). Sex was a factor in blood neutrophil concentration. Specifically, females had concentrations averaging 2000 cells per μl higher than males ($P < 0.0001$; Figure 6). We found no association between blood neutrophil concentration and race or ethnicity (comparing Black with non-Black, and Hispanic with non-Hispanic).

Differences between the present findings and those reported in the Manroe and Mouzinho publications^{1,2} are listed in the Table 1. All values are not strictly comparable since the Mouzinho study evaluated counts in neonates <1500 g, while this study was <36 weeks gestation. The Manroe and Mouzinho charts have narrower ranges of values than does this report. If neutrophilia is defined, using the present charts, as a value greater than the 95th percentile, it would be diagnosed more commonly using the Manroe or Mouzinho charts (Table 1).

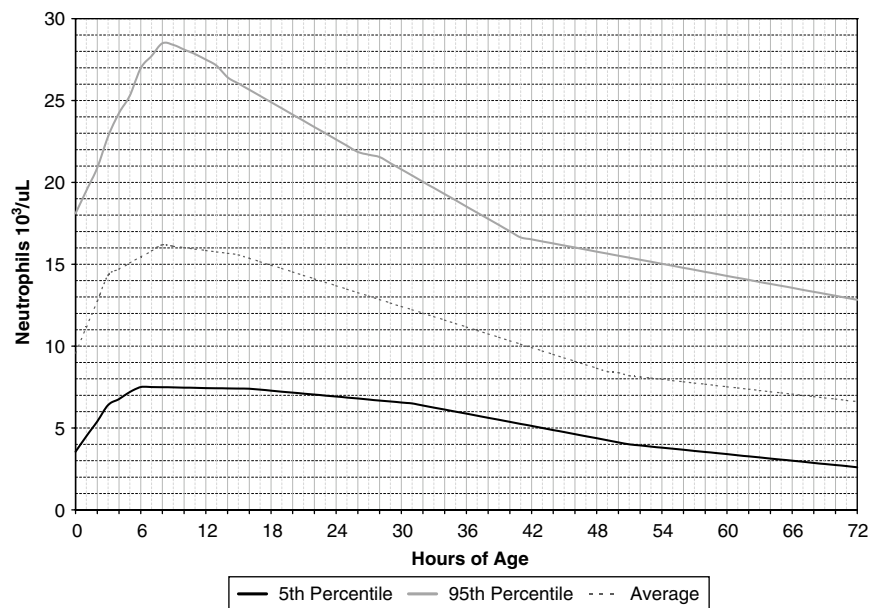


Figure 1 Neutrophils per μL of blood during the first 72 h after the birth of term and near-term (>36 weeks gestation) neonates. A total of 12 149 values were obtained for the analysis. The 5th percentile, the mean, and the 95th percentile values are shown.

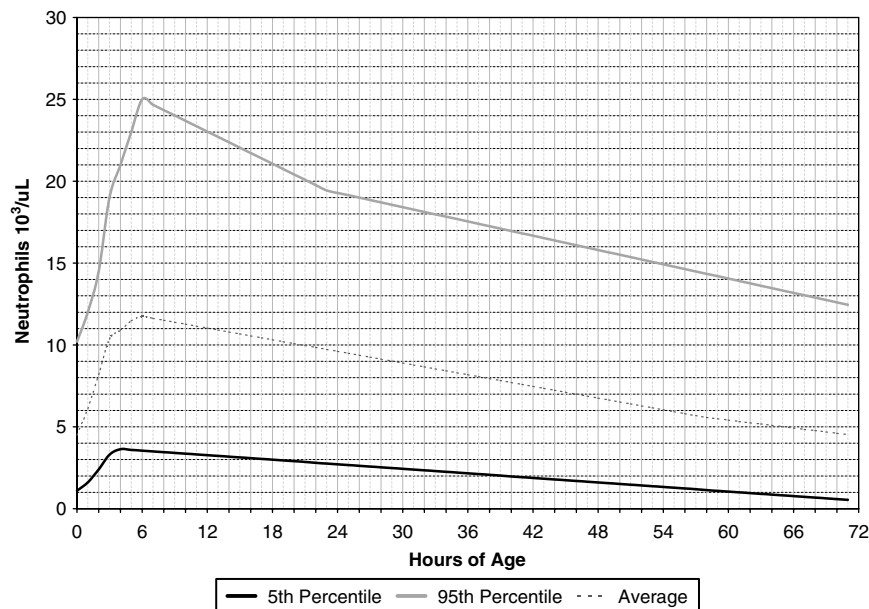


Figure 2 Neutrophils per μL of blood during the first 72 h after the birth of 28- to 36-week gestation preterm neonates. A total of 8896 values were obtained for the analysis. The 5th percentile, the mean, and the 95th percentile values are shown.

Discussion

The pioneering observations of the Rosenfeld group,^{1,2} in defining the normal ranges for blood neutrophil concentrations of neonates, are among the most widely used reports in the field of neonatal hematology.³ For this study, we used their 1979¹ and 1994² publications as templates, applying a modern blood cell analyzer and very large sample sizes to reexamine the range

of neutrophil counts of neonates. We made two additional modifications in approach; first, we did not include values from patients with certain diagnoses known to have a large proportion of either abnormally high neutrophil counts (Down syndrome,¹¹ trisomy 18, 13,¹² early-onset bacterial sepsis¹³) or abnormally low neutrophil counts (pregnancy-induced hypertension,^{5,6} early-onset bacterial sepsis^{7,8} or congenital neutropenia^{9,10}). Second, rather

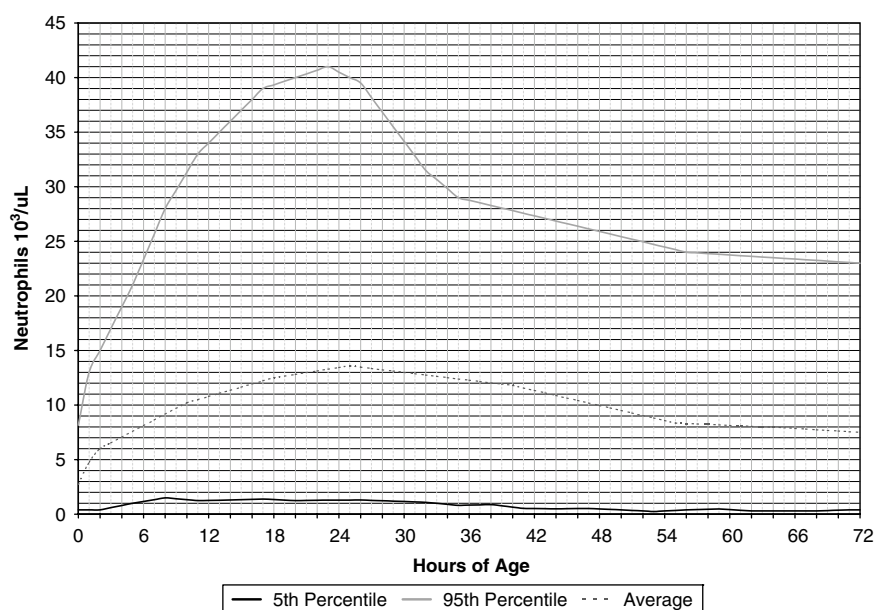


Figure 3 Neutrophils per μL of blood during the first 72 h after the birth of <28-week gestation preterm neonates. A total of 852 values were obtained for the analysis. The 5th percentile, the mean, and the 95th percentile values are shown.

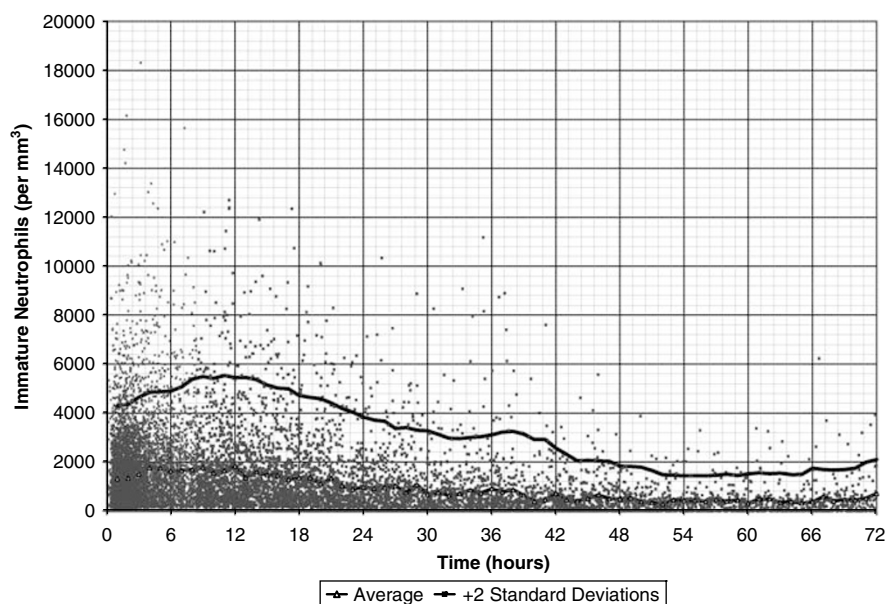


Figure 4 Immature neutrophils (band neutrophils plus metamyelocytes) per μL of blood during the first 72 h after the birth of term and nearterm (>36 weeks gestation) neonates. A total of 12 857 values were obtained for the analysis. The average and 2 s.d. above the mean value are shown.

than display the range of values as was done in the previous reports,^{1,2} we displayed the 5th percentile value, the mean value and the 95th percentile value. Our intent in making these modifications was to define an 'expected range' of blood neutrophil values for newborn infants in our health-care system. We are aware of no side-by-side comparisons, using blood of neonates, between the former methods for enumerating neutrophils^{1,2} and the more

modern methods (Beckman Coulter LH 750). However, using adult cells the modern methods give more reproducible results, owing to much larger sample sizes and more accurate identification methods.¹⁴

Our studies suggest a higher value for the upper limit of neutrophil counts than is indicated by the Manroe and Mouzhino charts.^{1,2} Neutrophilia could be defined using the

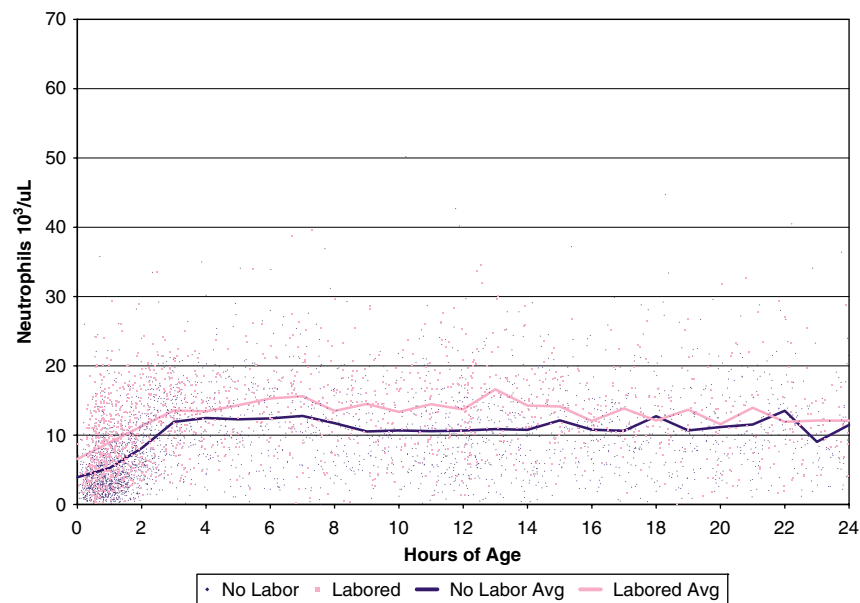


Figure 5 Neutrophils per μL of blood during the first 24 h after birth among 4080 neonates delivered by cesarean section with preceding labor vs without labor.

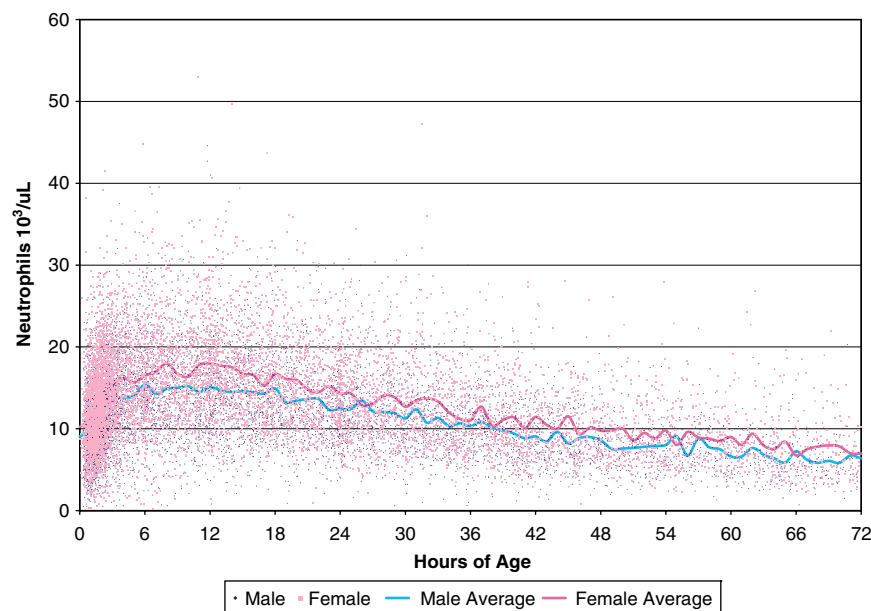


Figure 6 Neutrophils per μL of blood during the first 72 h after birth in 8950 males vs 6740 females.

present tables, at all gestational ages, by a neutrophil concentration exceeding about $13\,000\ \mu\text{L}^{-1}$ at 72 h of age. In contrast, the Manroe chart suggests that neutrophilia could be defined by counts above $7000\ \mu\text{L}^{-1}$. It is not clear why our counts had a much higher upper range. One possible explanation is the difference in methodology used (older vs newer methods for enumerating neutrophils). A second possible explanation is altitude. Dallas is 500 feet above sea level and the Intermountain Healthcare hospitals are at an average of 4800 feet above sea level. Carballo

et al.,¹⁵ Maynard *et al.*¹⁶ and Manroe¹⁷ reported that high-altitude results in higher blood neutrophil concentrations of newborn infants. They speculated that reduced oxygen delivery to the fetus is involved in the mechanism. However, the exact mechanistic explanation for this finding is not known. We suspect that altitude was indeed the main difference between the present range of neutrophil and that reported by Manroe¹ and Mouzinho,² because the Carballo¹⁵ and Maynard¹⁶ reports used the older methods, yet our ranges are similar to theirs.

Table 1 Comparison of the present results with those of the Manroe report¹ and the Mouzinho report²

	Upper limit at delivery (μl^{-1})	Lower limit at delivery (μl^{-1})	Time of peak (h)	Upper limit at peak (μl^{-1})	Lower limit at peak (μl^{-1})	Upper limit at 72–240 h (μl^{-1})	Lower limit at 72–240 h (μl^{-1})
<i>(A) The present report of neonates >36 weeks gestation compared with the Manroe report</i>							
Present	18 000	3500	8	28 500	7500	13 000	2700
Manroe	6000	1800	12	14 400	7800	7200	3000
<i>(B) The present report of neonates 28–36 weeks gestation compared with the Mouzinho report^a</i>							
Present	10 500	1000	6	25 000	3500	12 500	800
Mouzinho	6900	2100	12	14 400	8200	6000	1100
<i>(C) The present report of neonates <28 weeks gestation compared with the Mouzinho report^a</i>							
Present	8000	500	24	41 000	1500	15 300	1300
Mouzinho	6900	2100	12	14 400	8200	6000	1100

^aThe average gestational age in the Mouzinho was 29 weeks.

Speculation has arisen regarding the cause of the transient increase in blood neutrophil count in the first day following delivery.³ Our present data indicate that this is, in part, the result of labor, because among neonates delivered by cesarean section, those with preceding labor had considerably higher neutrophil concentrations than did those with no preceding labor. Labor does not explain the entire observation, however, since those with no labor still had the characteristic 'rollercoaster' shape of the curve over the first 72 h. Blood catecholamine concentrations increase after delivery as part of the transition to extrauterine life,^{18,19} and it is likely that epinephrine-stimulated demargination of neutrophils is involved in this early and transient increase in blood neutrophil concentration. Generally, neutrophil demargination occurs within minutes of epinephrine administration and remargination occurs in an hour or so.²⁰ Thus the pattern of blood neutrophil counts after delivery, as shown in Figures 1 and 2, does not precisely conform to an epinephrine-induced demargination study, because the return to baseline in Figures 1 and 2 takes several days. Perhaps a lengthy period of neutrophil demargination, due to several hours of epinephrine release, is responsible for this peak, or perhaps factors other than neutrophil demargination are involved. Since a 'left shift' does not generally occur in normal neonates following delivery,^{1–3} the peak is probably not the kinetic result of accelerated release of neutrophils from the marrow into the blood (which should increase the immature to total neutrophil proportion).

Neutropenia would be diagnosed similarly whether the Manroe and Mouzinho charts^{1,2} or the present charts were used. This is the case despite our efforts to exclude data from patients with a high likelihood of low neutrophil counts (such as neonates delivered to women with pregnancy-induced hypertension). In general, the previous and present charts would label a term/near-term neonate as neutropenic if the neutrophil count was below about $2500 \mu\text{l}^{-1}$

(after 72 h), and would label a preterm neonate as neutropenic if the count was below about $1000 \mu\text{l}^{-1}$.

One limitation of this study is a lack of accounting for the vascular source of each blood sample tested, because this was not recorded at the time of phlebotomy and is not part of our electronic records. Blood drawn from an artery tends to have a lower neutrophil concentration than blood drawn simultaneously from a vein or capillary source.^{21–23} This difference is likely related to the pulsatile flow of arterial blood, 'pushing' the larger cells away from the rapidly flowing central stream and toward the arterial walls. Because of this, the central stream of arterial blood has a lower concentration of neutrophils. The limitation of not accounting for the vascular source of the blood samples also exists for all previous studies attempting to define the normal range of neutrophil concentrations in neonates. We are certain that some of the 30 354 CBCs used in this report were from arterial blood, specifically from umbilical artery catheters. We speculate that fewer than 10% of the samples in this report were from arterial sources, but we have no means of knowing this precisely or of knowing which samples these were. We speculate further that if those samples could be identified and the values removed from the figures of this paper, it would have a relatively small effect; slightly raising some of the values within each figure.

Before the Dallas reports,^{1,2,7,13,17} it was not clear that the CBC was a particularly useful test in neonatology. However, since those publications the CBC has become a commonly performed test in the neonatal intensive care unit, and one conveying a multiplicity of information not otherwise available.³ On that basis, we judge that efforts to refine the interpretability of CBCs from neonatal intensive care unit patients are worthwhile,²⁴ and we maintain that our present update of the landmark papers of the Dallas group is a useful step in that direction. The publications by Carballo *et al.*¹⁵ and Maynard *et al.*¹⁶ recommended that neonatal centers at high

altitude report their neonatal neutrophil counts. We have done so, with a very large sample size, and our report supports their conclusion that neonates at high altitude have higher upper range values for neutrophil counts during the first 10 days of life. As they speculated,^{15,16} using the Manroe¹ or Mouzinho² chart for neonates at high altitude will lead to inaccurately labeling many who actually have a normal blood neutrophil concentration as having neutrophilia.

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